Characterization of base pair geometry

Xiang-Jun Lu,[a] & Wilma K. Olson[b]

[a]*Department of Biological Sciences, Columbia University, New York, NY 10027*

[b]*Department of Chemistry and Chemical Biology, Rutgers – The State University of New Jersey, Piscataway, NJ 08854*

Correspondence email: xiangjun@x3dna.org

Introduction

The interactions of the planar nitrogenous bases (A, C, G, T, and U) play a critical role in the three-dimensional organization of DNA and RNA. The relative spatial arrangement of a pair of associated bases can be rigorously quantified by six rigid-body parameters (Dickerson et al., 1989): three translational parameters called Shear, Stretch, and Stagger, and three rotational parameters denoted Buckle, Propeller (twist), and Opening (figure 1). The numerical values of these base pair parameters or the six step parameters used to describe the positioning of neighboring base pairs depend upon the choice of reference frame (Lu et al., 1999). The establishment of a standard base reference frame (Olson et al., 2001) and its implementation in 3DNA (Lu et al., 2003) and Curves+ (Lavery et al., 2009) has largely resolved discrepancies in the analysis of double-helical structures of Watson-Crick (WC) base pair.

The standard reference frame (Olson et al., 2001) is base centric, defined with respect to an 'idealized', perfectly planar WC pair where all six parameters are null. As noted in the classic Dickerson B-DNA dodecamer (Drew et al., 1981), WC pairs are normally non-planar, with Propeller (and Buckle) significantly different from zero. Notably, Propeller in right-handed DNA double helices has a mean value of around -12° (Dickerson et al., 1989; Olson et al., 2001), and its persistence has been rationalized in terms of the increased base-stacking interactions found in these structures (Calladine, 1982; Levitt, 1978). More generally, among the six base pair parameters, Buckle, Propeller, and Stagger describe the *non-planarity* of a pair: Buckle and Propeller render the two bases *non-parallel*, whilst Stagger leads to a vertical separation. On the other hand, Shear, Stretch, and Opening are critical for characterizing different types of non-WC pairs (Lu et al., 2015; Lu et al., 2003). For example, the G–U wobble pair is characterized by a Shear of -2.2 Å; if counted the other way around, i.e., as U–G, the Shear value is +2.2 Å instead.

3DNA (Lu et al., 2003) uses the standard base reference frame (Olson et al., 2001) to calculate six *local* base pair parameters with its 'analyze' routine. The parameters unambiguously characterize any base pair, be it WC, G–U wobble, or non-canonical (e.g., a Hoogsteen pair). Conversely, given the six parameters and base sequence, the 3DNA 'rebuild' routine rigorously reconstructs the spatial disposition of the two interacting bases. This reversibility is one of the unique features of 3DNA, applicable to pairs of bases in both DNA and RNA.

Simple base pair parameters

Although the six *local* base pair parameters are rigorous and serve a good purpose, their numerical values can be cryptic for non-canonical pairs, most notably the values of Buckle and Propeller. Two recent CCN articles (Richardson, 2015a, 2015b) emphasized the importance of base

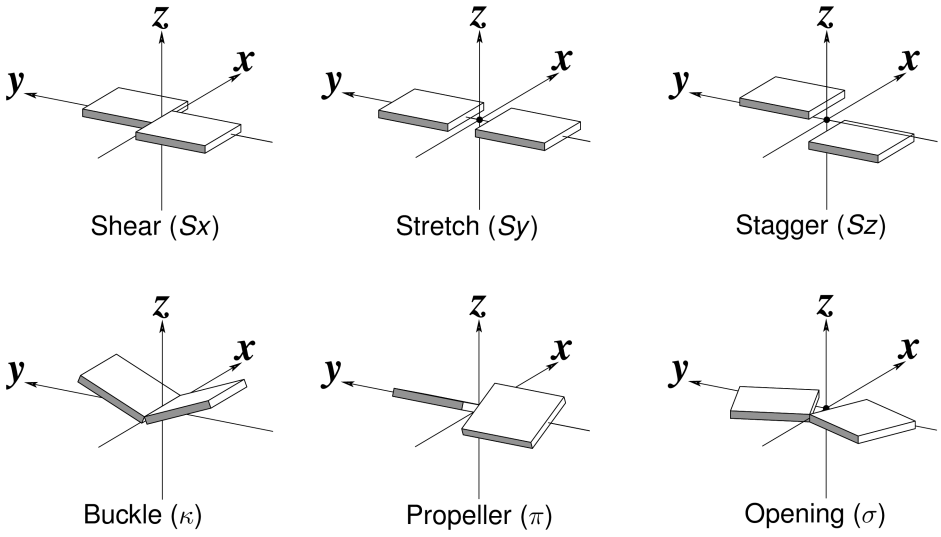


Figure 1: Schematic diagrams of the six rigid-body parameters commonly used for the characterization of base pair geometry.

pair non-planarity at biologically significant positions in high-resolution nucleic acid structures (e.g., functional binding sites) and the need to account correctly for this non-planarity in deriving models of DNA and RNA based on low-resolution data. The account of two of the classic measures of base pair deformations (figure 1) given in the article — "the 'propeller-twist' torsion around a line joining the two bases, and the 'buckle' angle of their bend across the line of base pair H-bonds" — prompted us to derive a new set of six *simple* parameters for a complete qualitative description of base pair geometry. The term '*Simple*' is used because the parameters are more intuitive for non-canonical pairs (figure 2), and to differentiate them from the existing *local* base pair parameters. The *simple* parameters have been implemented in both the 3DNA 'analyze' routine and the new DSSR software (Lu et al., 2015).

Detailed definitions of the *simple* base pair parameters, with worked examples, can be found on the 3DNA homepage (x3dna.org). The key differences between the *simple* and *local* parameters lie in the definition of the base pair coordinate frame and in the description of angular parameters. The *simple* treatment uses the locations of atoms on the interacting purine (R) and pyrimidine (Y) bases, either RC8/YC6 (default) or RN9/YN1, as the (long) ***y***-axis of the pair, corrected to be orthogonal with the ***z***-axis. The ***z***-axis is the average of the *z*-axes of the two bases, following the definition of the corresponding *local* base pair axis and taking the anti-parallel direction into consideration (e.g., in WC pairs). The (short) ***x***-axis fulfills the right-handed rule. The three translational parameters (figure 1) are simply the projections of the vector linking the origins of the two base-reference frames onto the respective ***x***-, ***y***- and ***z***-axes, similar to the definition of the corresponding *local* base pair parameters. The three *simple* angular parameters, however, are defined as 'torsion' angles: Buckle as the 'torsion' angle of the *z*-axes of the two bases with respect to the short base pair axis, Propeller as the 'torsion' angle of the two *z*-axes with respect to the long base pair axis, and Opening as the 'torsion' angle of the two *y*-axes with respect to the base pair ***z***-axis. While the *simple* base pair parameters of WC pairs closely

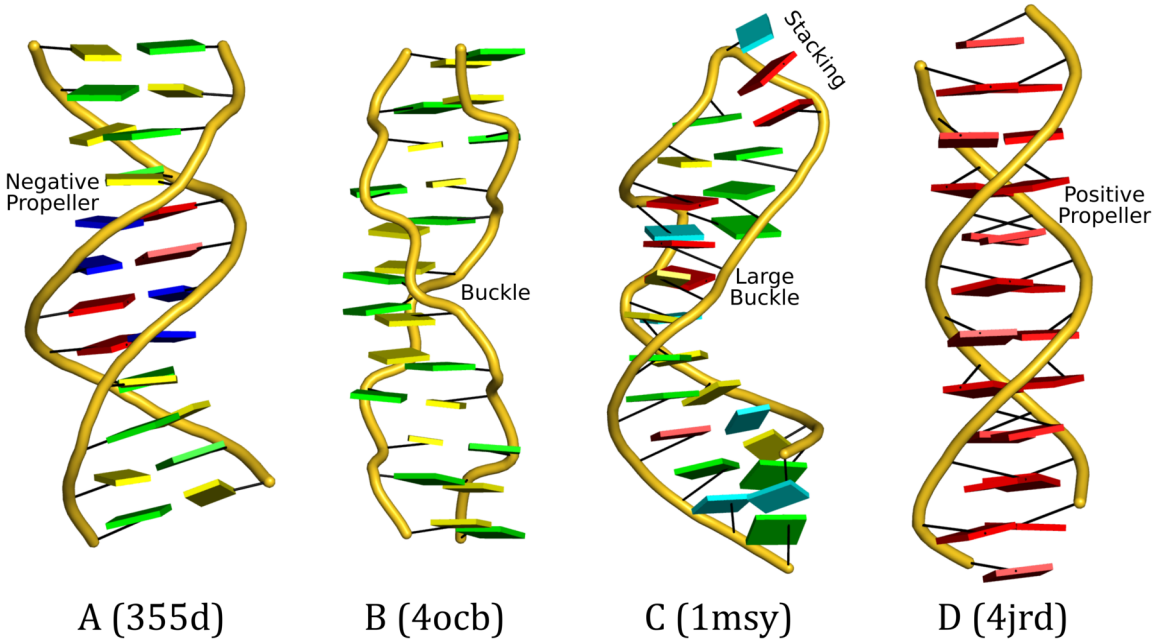


Figure 2: DSSR-introduced cartoon-block representations of DNA and RNA structures that combine PyMOL cartoon schematics with color-coded rectangular base blocks: A, red; C, yellow; G, green; T, blue; and U, cyan. (A) The Dickerson B-DNA dodecamer solved at 1.4-Å resolution [PDB id: 355d (Shui et al., 1998)], with significant negative Propeller. (B) The Z-DNA dodecamer [PDB id: 4ocb (Luo et al., 2014)], with virtually co-planar C–G pairs at the ends, and noticeable Buckle in the middle. (C) The GUAA tetraloop mutant of the sarcin/ricin domain from *E. coli* 23 S rRNA [PDB id: 1msy (Correll et al., 2003)], with large Buckle in the A+C pair, and base-stacking interactions of UAA in the GUAA tetraloop (upper-right corner). (D) The parallel double-stranded poly(A) RNA helix [PDB id: 4jrd (Safaee et al., 2013)], with up to +14° Propeller. The simple, informative cartoon-block representations facilitate understanding of the base interactions in small to mid-sized nucleic acid structures like these. The base identity, pairing geometry, and stacking interactions are obvious.

resemble the *local* base pair parameters, the two sets can differ significantly for non-canonical pairs.

Cartoon-block representations

The new DSSR component of 3DNA (Lu et al., 2015) includes an easy way to create highly effective cartoon-block representations that showcase base pair geometry and base-stacking interactions (figure 2). DSSR has been tested against all nucleic-acid-containing structures in the Protein Data Bank (PDB), and works with atomic coordinate files in either PDB or PDBx/mmCIF format. Figure 2 illustrates the arrangements of bases in four representative high-resolution X-ray crystal structures. The following command generates the script, named '`355d.pml`', needed to construct figure 2A (the Dickerson B-DNA dodecamer solved at 1.4 Å resolution, PDB id: 355d)within PyMOL:

```
x3dna-dssr -i=355d.pdb -o=355d.pml\
    --cartoon-block
```

Once '`355d.pml`' is loaded into PyMOL, the commands '`orient; turn z, -90; ray`' set the structure in the most extended view vertically and perform the ray-tracing needed to obtain the high-resolution PNG image. Similar commands apply to the other three cases. Details on reproducing the illustrated images, and outputs (including the *simple* base pair parameters) of DSSR and the 3DNA 'analyze' program for the four structures, are available at: x3dna.org/highlights/CCN-on-base-pair-geometry.

In addition to the default style shown in figure 2, DSSR provides several variations of the cartoon-block representation of planar, aromatic bases. For example, one can change both the thickness and the sizes of the blocks, shade the minor-groove edges of the bases, and represent a WC pair as a single long rectangular base pair block. One can also orient a structure in a specific base or base pair reference frame for easy comparison, and can attach the helical axes to an array of stacked base pairs. See the x3dna.org for more examples.

Summary

Base pair geometry can be described in different ways. The existing set of six *local* base pair parameters in 3DNA is mathematically rigorous, allowing for an unambiguous characterization of any pair of interacting bases and serving as input for exact model building. The new set of six *simple* base pair parameters described herein provides a more intuitive interpretation of intra-base pair structural variations, especially for the out-of-plane Buckle and Propeller distortions of non-canonical base pairs. The two sets of base pair parameters complement one another and serve different audiences and/or purposes. Numerical values of both sets of parameters are readily obtained with 3DNA. Moreover, the DSSR component of 3DNA provides highly effective cartoon-block representations of the base interactions within a nucleic acid structure.

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Note added in proof

The DSSR cartoon-block representation (figure 2) has been integrated into PyMOL via the 'dssr_block' command (http://pymolwiki.org/index.php/Dssr_block).